Fardeen Bablu

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EDUCATION

Vanderbilt University, Nashville, TN **May 2026**
Bachelor of Science in Computer Science, Mathematics, Public Health
Activities: Head Resident Advisor, Entrepreneurship Society, First Generation VU, Assoc. of Bengali Students

SKILLS

Programming: Rust, Go, C++, Python, TypeScript, Swift, SQL
Libraries: Actix-web, Tauri, Svelte, SwiftUI, React Native, Firebase, PyTorch, Qt6
Tools: Git, Linux, Docker, PostgreSQL/PostGIS, SQLite, DuckDB, CMake

EXPERIENCE

Research Engineer **Nashville, TN**
Vanderbilt Institute of National Security, Wicked Problems Lab Feb. 2026 – Dec. 2026
- Led a post-quantum TLS measurement platform over 33,069 .gov, financial, and university domains (Python, Go).
- Found 3,834 ML-KEM homepages whose login, checkout, or API endpoint still negotiated only classical key exchange.
- Wrote a Go prober (200 concurrent handshakes, native X25519MLKEM768) replacing a subprocess-OpenSSL scanner.
- Enforced in code and CI that timeouts record as Unknown, never as no-PQ; 1,000+ tests, ruff, and go vet gate every PR.

Teaching Assistant **Nashville, TN**
Vanderbilt University, Department of Computer Science Jan. 2025 – Dec. 2025
- Held 12 hours/week of office hours for 50+ students on Java fundamentals, debugging, OOP, and algorithm design.
- Graded 10+ coding assignments per cycle through GitHub and the course autograder.
- Supported 8 Software Engineering course teams; advised on design patterns, testing, and full-stack development.

Software Engineering Intern **Washington, DC**
Jenner & Block LLP June 2024 – Aug. 2024
- Built Spark, a Swift/SwiftUI iOS intranet prototype (people, news, events, search) over the firm's public data.
- Led a digital transformation initiative for 50+ attorneys, automating new-hire onboarding workflows.
- Configured Cisco Meraki access points across the office and helped integrate 5+ enterprise applications.

PROJECTS

Vectr - Pre-Scene Intelligence for EMTs - Rust, PostGIS, Tauri, Svelte - Case Study | Demo Jan. 2026 – May 2026
- Built the Rust/Actix-web + PostGIS backend: 62k lines, 68 handlers, 14 migrations, 1,600+ tests, 6-job CI.
- Designed an on-device clinical pipeline: 101 Tauri commands, local Whisper speech-to-text, NEMSIS v3.5.1 export.
- Scored helicopter landing zones from Overture and USGS data; field-tested on a 3-day Greene County (TN) EMS ride-along.

HUSH-MESH - Convoy-Protection Simulation - Python, TypeScript - Pitch | GitHub Oct. 2025 – June 2026
- 2nd place ($15K) at the Amazon Mission Autonomy Hackathon 2025: an A* mission planner over MQTT edge agents.
- Built the React simulation of convoy, drones, and threats; rewrote it post-hackathon, fixing 5 documented bugs.
- Fixed a cross-thread asyncio bug that silently discarded every planner recommendation; added CI (ruff, pytest, TS build).

Niyah - Real-money Focus App (iOS) - Swift, Firebase, Stripe, Plaid - niyah.live | GitHub Jan. 2026 – July 2026
- Shipped TestFlight builds whose native Swift Screen Time extensions earned Apple's FamilyControls entitlement.
- Built Apple Pay + Stripe Connect checkout and Plaid payouts over 8,800 lines of Cloud Functions with a ledger invariant.
- Led the React Native + Expo build: 1,000+ tests, 4 CI workflows including a money-path suite, Sentry observability.

QuizLITE - Flashcard Desktop App - C++17, SQLite, Qt6, CMake, GTest - GitHub Sept. 2024
- Built a flashcard app with 3 adaptive modes prioritizing low-accuracy questions via spaced repetition.
- Architected the SQLite backend with singleton-pattern CRUD operations and per-question accuracy tracking.
- Configured GitHub Actions with 35+ GoogleTest cases across Ubuntu and macOS, plus clang-format linting.

AWARDS

Amazon Mission Autonomy Hackathon 2025 - 2nd Place ($15K Prize) - Pitch | GitHub Nashville, TN
HUSH-MESH, a convoy-protection simulation with an A* mission planner over MQTT edge agents (see Projects).

Vanderbilt X Ironsite Hackathon 2025 - 3rd Place ($2.5K Prize) - Pitch | GitHub Nashville, TN
Led the idea and pitch for a 4-person team's occlusion-aware object tracking for construction-site video.

Google Solutions Challenge 2025 - Demo Day - Pitch New York, NY
Gemicast, a real-time outage-risk dashboard with rule-based risk alerts, inspired by recent Tennessee floods.